



Retriever Performance Report

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Dit rapport vergelijkt 4 retrievers voor retrieval-augmented generation. Het geeft een samenvatting van de corpus- en queryset, vergelijkt de rankingkwaliteit en runtime, en toont vervolgens de configuratie en voorbeeldresultaten van elke retriever. Gebruik de vergelijkingsectie om te bepalen welke retriever het meest waarschijnlijk nuttig bewijs in het contextvenster van de generator plaatst.

De gesynthetiseerde retrieval-dataset is ontworpen voor het evalueren van retrieval-methoden binnen Retrieval-Augmented Generation (RAG), en bevat een corpus van technische tekstdocumenten, zoals codevoorbeelden, uitleg over frameworks en implementatie-instructies, gericht op PHP en Symfony. De dataset bevat ook synthetisch gegenereerde gebruikersvragen die op basis van technische problemen en oplossingen zijn opgesteld, zoals hoe bepaalde functionaliteiten moeten worden geïmplementeerd, geconfigureerd of opgelost. Deze vragen zijn kort, technisch en gericht op specifieke componenten of problemen binnen een development-context. De dataset wordt gebruikt om de efficiëntie en nauwkeurigheid van retrieval-systemen te meten, door relevante documenten te identificeren op basis van de gestelde vragen. De relevantiebeoordelingen tussen vragen en documenten helpen bij het beoordelen van hoe goed een retrieval-model de juiste informatie kan vinden, op basis van technische context en probleemoplossing, in plaats van slechts woordoverlapping of onderwerpvergelijking.

Machine Summary

Machine Summary

System	Linux
Processor	x86_64
Cores	32
Ram (Gb)	377.56
Python Version	3.12.3
Cuda	Available (2 GPU(s))
Gpu Name	NVIDIA RTX A5000

Corpus Summary

Corpus and Queries

Documents	2,594
Documents With Title	0
Avg Document Chars	695.1766
Min Document Chars	5
Max Document Chars	1,000
Queries	100
Avg Query Chars	21.3
Min Query Chars	12
Max Query Chars	32

Sample Query

path make absolute
symfony scheduler stateful
event dispatcher setup
cache pool namespace
symfony bundle testing

Retriever Comparison

Retrieval and Rerank Metrics

Retriever	NDCG@10	Recall@10
BM25Retriever (bm25)	0.6414	0.6275
BM25Retriever (bm25) + rerank	0.7345	0.691
DenseRetrieverSentenceBert (dense)	0.7369	0.747
DenseRetrieverSentenceBert (dense) + rerank	0.7995	0.781
HybridRetriever (hybrid)	0.7231	0.725
HybridRetriever (hybrid) + rerank	0.7987	0.779
LinearRagRetriever (linear)	0.752	0.7635
LinearRagRetriever (linear) + rerank	0.8078	0.795

Chart Colors

Series
Base retrieval
Rerank



Uitleg metrics: Deze scores evalueren de retrievelfase voordat het taalmodel een antwoord genereert. Binnen een RAG-pipeline betekent betere retrieval dat de generator relevanter en beter gerankt bewijs ontvangt en minder hoeft te vertrouwen op niet-onderbouwde modelkennis.

Recall@k meet dekking: hoeveel relevante documenten zijn gevonden binnen de top k. **Precision@k** meet focus: hoeveel van de top k daadwerkelijk relevant is in plaats van afleidende context. **MAP@k** beloont relevante documenten die consistent vroeg verschijnen over meerdere queries heen. **NDCG@k** beloont zowel de rangorde als graduele relevantie en is daarom vaak de beste enkele indicator voor de vraag of het sterkste bewijs als eerste de prompt bereikt.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de hoogste score te kiezen, maar om te bepalen welke retriever het beste past bij het doel van de RAG-toepassing. Een hoge Recall@k is vooral belangrijk wanneer het systeem zo min mogelijk relevante informatie mag missen, bijvoorbeeld bij juridische of ondersteunende zoekvragen. Een hoge Precision@k is belangrijk wanneer het contextvenster beperkt is en irrelevante passages de generator kunnen afleiden. Wanneer reranking duidelijk betere NDCG- of MAP-scores geeft, betekent dit dat de juiste documenten niet alleen gevonden worden, maar ook hoger in de ranglijst komen te staan. Dat is praktisch waardevol omdat een taalmodel meestal sterker leunt op de eerste passages in de context. Runtime moet daarnaast worden meegewogen: een iets lagere score kan acceptabel zijn wanneer de retriever veel sneller is of eenvoudiger te onderhouden blijft.

Practical Limitations and Runtime Trade-offs

Retriever	Query Time	Queries/Sec	Index Time	Document/Sec	Storage
BM25Retriever (bm25)	1.0976s	91.1058	10.3464s	250.7148	2.11MB
BM25Retriever (bm25) + rerank	10.0377s	9.9625	10.3464s	250.7148	2.11MB
DenseRetrieverSentenceBert (dense)	0.0386s	2587.7913	38.7569s	66.93	10.62MB
DenseRetrieverSentenceBert (dense) + rerank	8.7403s	11.4434	38.7569s	66.93	10.62MB
HybridRetriever (hybrid)	1.1381s	87.8658	49.1034s	52.8273	12.73MB
HybridRetriever (hybrid) + rerank	9.7997s	10.2061	49.1034s	52.8273	12.73MB
LinearRagRetriever (linear)	22.7802s	4.3898	60.0930s	43.1664	2.29GB
LinearRagRetriever (linear) + rerank	31.4634s	3.1786	60.0930s	43.1664	2.29GB

Quality versus Query Throughput

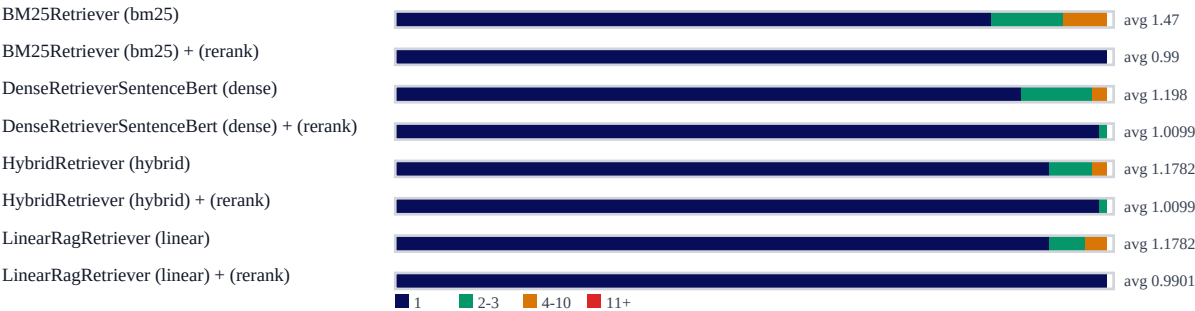


Uitleg runtime en throughput: Deze resultaten evalueren de praktische prestaties van de retrievelfase naast retrievalkwaliteit. Query time meet hoeveel tijd nodig is om queries te verwerken, terwijl index time aangeeft hoeveel tijd nodig is om de volledige index op te bouwen. Queries/sec en documents/sec meten de throughput van het systeem en geven inzicht in schaalbaarheid binnen grotere RAG-omgevingen. Storage MB toont hoeveel opslagruimte de retrievalindex gebruikt, wat belangrijk kan zijn binnen productieomgevingen waar infrastructuurkosten en geheugenverbruik een rol spelen. De grafiek combineert retrievalkwaliteit met throughput door Recall@10 af te zetten tegen queries per seconde. Hogere posities betekenen betere retrievaldekking, terwijl posities verder naar rechts wijzen op hogere snelheid en lagere latency.

Praktische interpretatie: Gebruik deze resultaten niet alleen om de snelste of meest accurate retriever te kiezen, maar om te bepalen welke retrievalarchitectuur het beste aansluit bij de praktische eisen van de toepassing. Een retriever rechtsboven in de grafiek combineert hoge retrievalkwaliteit met hoge throughput en vormt daardoor meestal de beste algemene balans voor real-time RAG-systemen. Een retriever kan echter bewust meer querytijd gebruiken om betere ranking of hogere retrievaldekking te behalen. In productieomgevingen moet daarom rekening worden gehouden met latency, schaalbaarheid, hardwarekosten en onderhoudscomplexiteit. Een iets lagere retrievalscore kan acceptabel zijn wanneer een systeem aanzienlijk sneller, goedkoper of eenvoudiger schaalbaar blijft.

Rank Distribution

Retriever	Queries	Rank 1	Rank 2-3	Rank 4-10	Rank 11+	Avg First Rank
BM25Retriever (bm25)	100	83	10	6	0	1.47
BM25Retriever (bm25) + (rerank)	100	99	0	0	0	0.99
DenseRetrieverSentenceBert (dense)	101	88	10	2	0	1.198
DenseRetrieverSentenceBert (dense) + (rerank)	101	99	1	0	0	1.0099
HybridRetriever (hybrid)	101	92	6	2	0	1.1782
HybridRetriever (hybrid) + (rerank)	101	99	1	0	0	1.0099
LinearRagRetriever (linear)	101	92	5	3	0	1.1782
LinearRagRetriever (linear) + (rerank)	101	100	0	0	0	0.9901

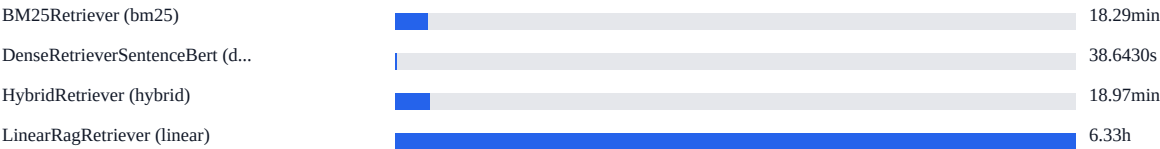


Uitleg rank distribution: Rank distribution laat zien op welke positie het eerste relevante document verschijnt voor iedere query. In plaats van alleen een algemene retrievalscore te tonen, maakt deze verdeling zichtbaar hoe snel bruikbare informatie beschikbaar komt voor de RAG-pipeline. Rank 1 betekent dat direct het eerste resultaat relevant is. Rank 2-3 betekent dat relevante informatie nog steeds zeer vroeg wordt gevonden. Rank 4-10 geeft aan dat de retriever dieper in de ranking moet zoeken voordat bruikbaar bewijs verschijnt. Rank 11+ betekent dat relevante informatie pas laat wordt gevonden, waardoor de kans groter wordt dat het taalmodel minder sterke of minder gefocuste context ontvangt. De gemiddelde first rank vat samen hoe vroeg het eerste relevante document gemiddeld verschijnt; lagere waarden duiden op sterkere rankingprestaties.

Praktische interpretatie: Een retriever met veel Rank 1 en Rank 2-3 resultaten levert doorgaans directer bruikbaar bewijs voor generatie. Dit is vooral waardevol wanneer het contextvenster beperkt is of wanneer de generator sterk afhankelijk is van de eerste passages in de prompt. Een groter aandeel Rank 4-10 kan nog steeds bruikbaar zijn, maar betekent dat relevante informatie minder prominent wordt aangeboden. Een groot Rank 11+ aandeel wijst vaak op zwakkere rankingkwaliteit of op een grotere behoefte aan reranking, hybride retrieval of betere queryverwerking. Rank distribution helpt daarom niet alleen te bepalen of relevante documenten worden gevonden, maar vooral hoe snel en hoe zichtbaar deze documenten voor de generator beschikbaar zijn.

Scalability Estimates

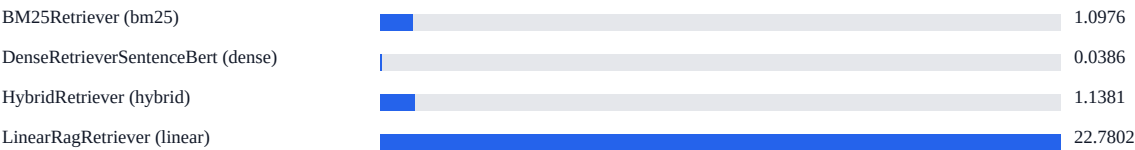
Retriever	Target Docs	Est. Index Time	Target Queries	Est. Query Time	Est. Queries/Sec	Est. Storage
BM25Retriever (bm25)	1,000,000	1.11h	100,000	18.29min	91.1058	813.23MB
DenseRetrieverSentenceBert (dense)	1,000,000	4.15h	100,000	38.6430s	2587.7913	4.00GB
HybridRetriever (hybrid)	1,000,000	5.26h	100,000	18.97min	87.8658	4.79GB
LinearRagRetriever (linear)	1,000,000	6.44h	100,000	6.33h	4.3898	881.85GB



Total Index Time (seconds) (2594 docs)



Total Query Time (seconds) (100 queries)

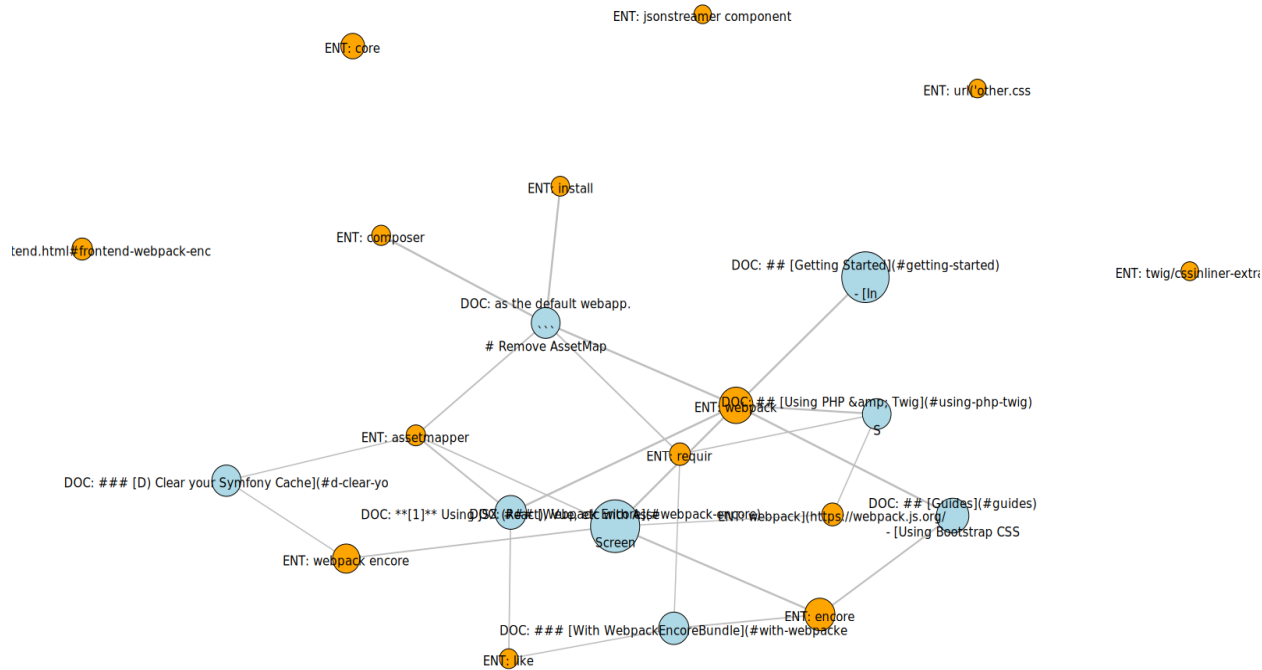


Rerank Time (seconds) (20 - docs / 100 - queries)



Query Graph - LinearRagRetriever (linear)

Query: webpack encore cdn



Retrieved Documents

Rank	Document	Score	Text
1	657d9e76-c5fb-454a-b6c3-e8574a48b26c	0.0164	### [Webpack Encore](#webpack-encore) Screencast Do you prefer video tutorials? Check out the [Webpack Encore screencast series...
2	e9563c4d-fc97-4a21-8397-149ded4f7458	0.0161	## [Getting Started](#getting-started) - [Installation](installation.html) - [Using Webpack Encore](simple-example.html) ## [Ad...
3	3896fb45-6376-4cdf-987c-63f384f3fa14	0.0159	## [Guides](#guides) - [Using Bootstrap CSS & JS](bootstrap.html) - [jQuery and Legacy Applications](legacy-applications.ht...
4	38e1c5f2-ef8d-4104-b73e-a8b100d7029d	0.0156	**[1]** Using JSX (React), Vue, etc with AssetMapper is possible, but you'll need to use their native tools for pre-compilation...
5	5a2c522a-a5a3-40ae-9aab-72c6b8a8dc10	0.0154	### [With WebpackEncoreBundle](#with-webpackencorebundle) If you're using Webpack Encore, the recipe will also update your `web...

6	9bcb2b7a-a36c-4354-a5a1-a67c54c25450	0.0152	### [D] Clear your Symfony Cache](#d-clear-your-symfony-cache) Make sure you clear and warm-up your Symfony cache: `` \$ APP_EN...
7	a50a3430-d27e-4632-9405-b3b1da42528c	0.0149	as the default webapp. `` # Remove AssetMapper & Turbo/Stimulus temporarily \$ composer remove symfony/ux-turbo symfony/asset-m...
8	ef5ecd5d-9204-45ca-969a-834af8c79494	0.0147	## [Using PHP & Twig](#using-php-twig) Symfony comes with two powerful options to help you build a modern and fast frontend...

BM25Retriever (bm25)

Configuration

Retriever Type	bm25
Topk	20
Language	english
Index Name	raccoon-bm25-test
Content Field	content
Metadata Field	metadata
Refresh On Write	yes
Timeout	120

Result Summary

Queries	100
Total Results	2,000
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	3.1599
Avg Score	9.1289
Max Score	24.6323

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.7867	0.636	0.5747	0.6414	0.6896
MAP	0.1758	0.3465	0.4359	0.4939	0.5276
Recall	0.1758	0.3682	0.4975	0.6275	0.748
Precision	0.83	0.5933	0.484	0.307	0.1835

Retriever Metrics

Index Time / Indexing / Time In Seconds	10.3464
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Docs Per Second	250.7148
Index Time / Indexing / Backend	elasticsearch
Index Time / Indexing / Storage Size Bytes	2,211,979
Index Time / Indexing / Storage Size Mb	2.1095
Query Time / Search / Time In Seconds	1.0976
Query Time / Search / Time Per Query In Seconds	0.011
Query Time / Search / Queries Per Second	91.1058
Query Time / Search / Queries	100

Query Time / Search / Top K	20
Query Time / Search / Backend	elasticsearch
Query Time / Search / Total Results	2,000
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	3.1599
Query Time / Search / Max Score	24.6323
Query Time / Search / Avg Score	9.1289

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	100
Total Pairs	2,000
Total Batches	200
Total Input Tokens	482,875
Total Wall Time Sec	8.9401
Avg Time Query Sec	0.0894
Avg Time Batch Sec	0.0447
Pairs Per Sec	223.71
Queries Per Sec	11.19
Tokens Per Sec	54012.35
Peak Gpu Memory Gb	1.21

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9467	0.7732	0.6757	0.7345	0.7579
MAP	0.2058	0.449	0.5591	0.6126	0.6289
Recall	0.2058	0.451	0.575	0.691	0.748
Precision	0.98	0.73	0.562	0.339	0.1835

Sample Top Results

Query q0: webpack encore cdn

1. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 22.6636

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2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 19.5482

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3. Document ef5ecd5d-9204-45ca-969a-834af8c79494 | score 16.3214

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Sample Top Rerank Results

Query q0: webpack encore cdn

1. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 1.0889

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2. Document 9bcb2b7a-a36c-4354-a5a1-a67c54c25450 | score 0.4868

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3. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 0.1893

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DenseRetrieverSentenceBert (dense)

Configuration

Retriever Type	dense
Topk	20
Model Id	snowflake/snowflake-arctic-embed-l-v2.0
Max Length	512
Device	cuda
Query Prompt Name	query
Passage Prompt Name	document
Normalize Embeddings	no
Batch Size	128
Corpus Chunk Size	50,000
Query Chunk Size	1,024
Show Progress Bar	yes

Result Summary

Queries	100
Total Results	2,000
Avg Results/Query	20
Min Results/Query	20

Max Results/Query	20
Min Score	0.1545
Avg Score	0.4102
Max Score	0.7742

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.8467	0.7243	0.6412	0.7369	0.7832
MAP	0.1858	0.4063	0.5066	0.5962	0.6308
Recall	0.1858	0.4223	0.5503	0.747	0.857
Precision	0.88	0.6867	0.54	0.367	0.211

Retriever Metrics

Index Time / Encoding / Time In Seconds	38.7569
Index Time / Encoding / Documents	2,594
Index Time / Encoding / Queries	100
Index Time / Encoding / Shards	1
Index Time / Encoding / Embedding Dim	1,024
Index Time / Encoding / Size In Bytes	11,136,732
Index Time / Encoding / Size In Mb	10.6208
Index Time / Encoding / Docs Per Second	66.93
Query Time / Search / Time In Seconds	0.0386
Query Time / Search / Time Per Query In Seconds	0.0004
Query Time / Search / Queries Per Second	2587.7913
Query Time / Search / Queries	100
Query Time / Search / Top K	20
Query Time / Search / Score Function	dot
Query Time / Search / Use Faiss	no
Query Time / Search / Backend	torch
Query Time / Search / Total Results	2,000
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.1545
Query Time / Search / Max Score	0.7742
Query Time / Search / Avg Score	0.4102

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	100
Total Pairs	2,000
Total Batches	200
Total Input Tokens	468,225
Total Wall Time Sec	8.7017
Avg Time Query Sec	0.087
Avg Time Batch Sec	0.0435
Pairs Per Sec	229.84
Queries Per Sec	11.49
Tokens Per Sec	53808.43
Peak Gpu Memory Gb	3.32

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9767	0.8297	0.7218	0.7995	0.8327
MAP	0.2098	0.4877	0.6107	0.6859	0.7101
Recall	0.2098	0.493	0.625	0.781	0.857
Precision	1	0.8	0.612	0.384	0.211

Sample Top Results

Query q0: webpack encore cdn

1. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 0.5299

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2. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 0.522

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3. Document 3896fb45-6376-4cdf-987c-63f384f3fa14 | score 0.4452

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Sample Top Rerank Results

Query q0: webpack encore cdn

1. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 1.0889

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2. Document 9bcb2b7a-a36c-4354-a5a1-a67c54c25450 | score 0.4868

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3. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 0.1893

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HybridRetriever (hybrid)

Configuration

Retriever Type	hybrid
K	60
Retrievers	bm25:0.3, dense:0.7

Result Summary

Queries	100
Total Results	2,000
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0087
Avg Score	0.0122
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.88	0.7111	0.6271	0.7231	0.7779
MAP	0.1938	0.3933	0.4838	0.573	0.614
Recall	0.1938	0.4103	0.5337	0.725	0.857
Precision	0.92	0.6667	0.522	0.356	0.211

Retriever Metrics

Query Time / Search / Time In Seconds	1.1381
Query Time / Search / Retriever Time In Seconds	1.1363
Query Time / Search / Rrf Time In Seconds	0.0018
Query Time / Search / Wrapper Time In Seconds	0.002
Query Time / Search / Execution Mode	sequential

Query Time / Search / Time Per Query In Seconds	0.0114
Query Time / Search / Queries Per Second	87.8658
Query Time / Search / Queries	100
Query Time / Search / Top K	20
Query Time / Search / Retrievers	2
Query Time / Search / Rrf K	60
Query Time / Search / Total Results	2,000
Query Time / Search / Avg Results Per Query	20
Query Time / Search / Min Results Per Query	20
Query Time / Search / Max Results Per Query	20
Query Time / Search / Min Score	0.0087
Query Time / Search / Max Score	0.0164
Query Time / Search / Avg Score	0.0122
Index Time / Indexing / Time In Seconds	49.1034
Index Time / Indexing / Documents	5,188
Index Time / Indexing / Docs Per Second	105.6547
Index Time / Indexing / Retrievers	2
Index Time / Indexing / Size In Mb	12.7303

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	100
Total Pairs	2,000
Total Batches	200
Total Input Tokens	468,225
Total Wall Time Sec	8.6616
Avg Time Query Sec	0.0866
Avg Time Batch Sec	0.0433
Pairs Per Sec	230.9
Queries Per Sec	11.55
Tokens Per Sec	54057.4
Peak Gpu Memory Gb	3.32

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9767	0.8297	0.7218	0.7987	0.8327
MAP	0.2098	0.4877	0.6107	0.6849	0.7098
Recall	0.2098	0.493	0.625	0.779	0.857
Precision	1	0.8	0.612	0.383	0.211

Sample Top Results

Query q0: webpack encore cdn

1. Document e9563c4d-fc97-4a21-8397-149ded4f7458 | score 0.0164

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1. Document 657d9e76-c5fb-454a-b6c3-e8574a48b26c | score 1.0889

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LinearRagRetriever (linear)

Configuration

Retriever Type	linear
Embedding Model Name	snowflake/snowflake-arctic-embed-l-v2.0
Spacy Model Name	nl_core_news_sm
Dataset Name	symphony_beir_lexical
Device	cuda
Max Seq Length	512
Embed Batch Size	128

Ner Batch Size	64
Retrieval Top K	20
Dense Candidate K	500
Bm25 Candidate K	500
Graph Candidate K	500
Local Graph Dense Seed K	200
Local Graph Bm25 Seed K	200
Local Graph Sentence Seed K	50
Local Graph Concept Seed K	50
Dense Rrf Weight	0
Bm25 Rrf Weight	0
Graph Rrf Weight	1
Rrf K	60
Min Concept Len	4
Max Concept Words	4
Min Concept Df	2
Max Concept Df Ratio	0.1
Ppr Damping	0.6
Ppr Max Result Docs	500
Export Query Graph	yes
Use Gpu For Spacy	yes

Result Summary

Queries	100
Total Results	2,000
Avg Results/Query	20
Min Results/Query	20
Max Results/Query	20
Min Score	0.0125
Avg Score	0.0143
Max Score	0.0164

Retrieval Metrics

Metric	@1	@3	@5	@10	@50	@100
NDCG	0.87	0.7349	0.6497	0.752	0.812	0.812
MAP	0.1938	0.4172	0.5126	0.617	0.6619	0.6619
Recall	0.1938	0.4297	0.5557	0.7635	0.9095	0.9095
Precision	0.92	0.6967	0.544	0.375	0.0896	0.0448

Retriever Metrics

Index Time / Indexing / Time In Seconds	60.093
Index Time / Indexing / Documents	2,594
Index Time / Indexing / Concepts	5,255

Index Time / Indexing / Sentences	26,480
Index Time / Indexing / Concepts Per Document	2.0258
Index Time / Indexing / Sentences Per Document	10.2082
Index Time / Indexing / Passage Concept Links	55,496
Index Time / Indexing / Concept Sentence Links	54,784
Index Time / Indexing / Sentence Passage Links	41,217
Index Time / Indexing / Embedding Dim	1,024
Index Time / Indexing / Size In Bytes	2,456,211,531
Index Time / Indexing / Size In Mb	2342.4259
Index Time / Indexing / Docs Per Second	43.1664
Index Time / Indexing / Cache Loaded	no
Index Time / Indexing / Cache Path	data/processed/symfony/symphony_beir_lexical/linear_rag_cache/snowf...
Index Time / Indexing / Embedding Model	snowflake/snowflake-arctic-embed-l-v2.0
Index Time / Indexing / Spacy Model	nl_core_news_sm
Index Time / Indexing / Backend	faiss
Index Time / Indexing / Phase Times / Model Load Time In Seconds	11.8014
Index Time / Indexing / Phase Times / Cache Lookup Time In Seconds	0.0004
Index Time / Indexing / Phase Times / Bm25 Build Time In Seconds	0.1643
Index Time / Indexing / Phase Times / Concept Extraction Time In Seconds	19.0209
Index Time / Indexing / Phase Times / Embedding Build Time In Seconds	21.8978
Index Time / Indexing / Phase Times / Cache Save Time In Seconds	5.9687
Query Time / Search / Time In Seconds	22.7802
Query Time / Search / Time Per Query In Seconds	0.2278
Query Time / Search / Queries Per Second	4.3898
Query Time / Search / Queries	100
Query Time / Search / Top K	20
Query Time / Search / Dense Candidate K	500

Reranker

Enabled	yes
Model Id	BAAI/bge-reranker-v2-m3
Top K	20
Batch Size	16
Device	cuda

Rerank Metrics

Total Queries	100
Total Pairs	2,000
Total Batches	200
Total Input Tokens	490,691
Total Wall Time Sec	8.6832
Avg Time Query Sec	0.0868
Avg Time Batch Sec	0.0434
Pairs Per Sec	230.33
Queries Per Sec	11.52
Tokens Per Sec	56510.23
Peak Gpu Memory Gb	4.38

Rerank Retrieval Metrics

Metric	@1	@3	@5	@10	@20
NDCG	0.9733	0.8278	0.7303	0.8078	0.8565
MAP	0.2098	0.489	0.626	0.6995	0.737
Recall	0.2098	0.493	0.641	0.795	0.9095
Precision	1	0.8	0.628	0.391	0.224

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